Printed Pages:01 **Sub Code: RCS702**

Paper Id: 2 3 1 3 9 8 **Roll No.** | | | | | | | | | | |

B.TECH.
(SEM VII) THEORY EXAMINATION 2022-23
ARTIFICIAL INTELLIGENCE

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *any Seven* questions in brief. 2 x7 = 14

(a) Explain the termcomputer vision in AI.
(b) Discuss the term heuristic search.
(c) Define the term Inference engine.
(d) What is a Search graph of a state space?
(e) Define natural language processing.
(f) Define conditional probability.
(g) List down some of the uninformed search techniques in state space search.
(h) Discuss applications of support vector machine in artificial intelligence.
(i) Differentiate between classification and regression technique.
(j) Explain the need of probabilistic reasoning in artificial intelligence.

SECTION B

2. Attempt any *three* of the following: 7x3=21

(a) What do you mean by AI? Explain the use of AI in various fields.
(b) Explain Hill Climbing algorithm in detail along with its limitations.
(c) What is prepositional logic? Define the various inference rules with the help of example.
(d) Define conditional probability and explain Naïve Bayes rule in detail.
(e) Explain the concept of pattern recognition along with suitable example.

SECTION C

3. Attempt any *one* part of the following: 7x1=7

(a) Discuss the elements of an agent and explain the characteristics of intelligent agent.
(b) What is the role of AI in the natural language processing? Illustrate the various phases in natural language processing.

4. Attempt any *one* part of the following: 7x1=7

(a) Discuss various searching techniques in AI.
(b) Explain the term heuristic function along with its benefits/uses in the field of AI.

5. Attempt any *one* part of the following: 7x1=7

(a) Elaborate the term regression along with comparison between linear regression and non-linear regression.
(b) Define the term reinforcement learning. Explain how passive reinforcement learning differ from active reinforcement learning.

6. Attempt any *one* part of the following: 7x1=7

(a) Explain the EM algorithm along with suitable example.
(b) Describe Bayesian networks and explain how Bayesian networks are a powerful representation for uncertainty knowledge.

7. Attempt any *one* part of the following: 7x1=7

(a) Explain k-means clustering with the help of suitable example in the field of pattern recognition.
(b) Explain the concept of Principle component analysis (PCA) in dimension reduction along with suitable example.

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